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B. TECH
(SEM VI) THEORY EXAMINATION 2022-23
REFRIGERATION AND AIR CONDITIONING

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

(a) What is the difference between heat pump and refrigerator?
(b) Define "refrigeration effect".
(c) What are the advantages of compound compression with intercooler over single stage compression?
(d) Draw and explain P-h Chart
(e) Write the name of components used in place of compressor in VAS.
(f) What are the desirable properties of an ideal refrigerant?
(g) Define sensible heat factor.
(h) What is Air stratification?
(i) Why the ducts are used in air conditioning system?
(j) Define fouling factor.

SECTION B

2. Attempt any *three* of the following: 10 x 3 = 30

(a) Enumerate the classification of air craft refrigeration system. Describe with a diagram the reduced ambient air cooling system.

(b) An ammonia refrigerating machine fitted with an expansion valve work between the temperature limits of - l0°C and 30°C. The vapour is 95% dry at the end isentropic compression and the fluid leaving the condenser is at 30°C. Assuming actual C.O.P. as 60% of the theoretical, calculate the kilograms of ice produced per kW hour at 0°C from water at 10°C. Latent heat of ice is 335 KJ/Kg. Ammonia has the following properties;

Temperature (°C)	Liquid Heat (KJ/Kg)	Latent Heat (KJ/Kg)	Liquid Entropy (KJ/Kg K)	Vapour Entropy (KJ/Kg K)
30	323.08	1145.8	1.2037	4.9842
-10	135.37	1297.68	0.5443	5.4770

(c) Differentiate between physical and thermodynamic properties of a refrigerant. Explain which more important giving specific examples are.

(d) Classify different types of air-conditioning system used. Draw a neat diagram of air-conditioning system required for winter season. Explain the working of different components in the circuit.

(e) What is the function of Cold storages and what is their importance in today's life? 2000 tonnes of potato are available at a temperature of 30 °C. It has to be preserved in a cold storage at a temperature of 2°C. How much refrigeration is necessary? If this refrigeration is to be obtained in 3 days, what should be the capacity of the plant?

SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

(a) The following data refers to a boot strap air cycle evaporative refrigeration system used for an aero-plane to take 20 tonnes of refrigeration load :
Ambient air temperature = 15°C, Ambient air pressure = 0.8 bar, Mach number of the flight

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= 1.2, Ram efficiency = 90%, Pressure of air bled off the main compressor = 4 bar, Pressure of air in the secondary compressor = 5 bar, Isentropic efficiency of the main compressor = 90%, Isentropic efficiency of the secondary compressor = 80%, Isentropic efficiency of the cooling turbine = 80%, Temperature of air leaving the first heat exchanger = 170°C, Temperature of air leaving the second heat exchanger = 155°C, Temperature of air leaving the evaporator = l00°C, Cabin temperature = 25°C, Cabin pressure = 1 bar.
Find; 1. Mass of air required to take the cabin load, 2. Power required for the refrigeration system, and 3. C.O.P. of the system.

(b) Explain, with a neat sketch, the working principle of boot-strap evaporative type of air refrigeration system with T-S diagram.

4. Attempt any *one* part of the following: 10 x 1 =10

(a) Explain two stage cascade refrigeration system with P-h and T-S diagram.

(b) For a vapour compression refrigeration system using R-22 as refrigerant, condenser outlet temperature is 40°C and the evaporator inlet temperature is - 20°C. In order to avoid flashing of the refrigerant, a liquid-suction vapour heat exchanger is provided where liquid is sub-cooled to 26°C. The refrigerant leaves the evaporator as saturated vapour. The compression process is isentropic. Find the power requirement and coefficient of performance if capacity of the system is 10 kW at -20°C. Shows the cycle on T-S and P-h diagrams. The specific heat of vapour refrigerant IS 1.03 KJ/Kg K. The properties are given below;

Sat. Temp (°C)	Pressure (bar)	Sp.volume (m^3/Kg)	Enthalpy (h_f) (KJ/Kg)	Enthalpy (h_g) (KJ/Kg)	Entropy (S_f) (KJ/KgK)	Entropy (S_g) (KJ/Kg)
-20	2.458	0.093	22.21	243.25	0.0908	0.9638
26	10.819	0.0218	79.74	260.64	0.2935	0.9014
40	15.489	0..0148	97.94	263.21	0.3563	0.8822

5. Attempt any *one* part of the following: 10 x 1 = 10

(a) In an absorption type refrigerator, the heat is supplied to NH3 generator by condensing steam at 2 bar and 90% dry. The temperature in the refrigerator is to be maintained at-5° C. Find the maximum C.O.P. possible, if the refrigeration load is 20 tonnes and actual C.O.P. is 70% of the maximum C.O.P, find the mass of steam required per hour. Take temperature of the atmosphere as 30° C.

(b) What is the basic function of a compressor in vapour compression refrigeration system? How this function is achieved in vapour absorption refrigeration system?

6. Attempt any *one* part of the following: 10 x 1 = 10

(a) Room air having a DBT of 40°C and WBT of 25°C is cooled through sensible cooling process upto a temperature of 25^0C show it on a psychometric chart and determine the amount of heat removed (in KJ/kg of dry air).

(b) Discuss, briefly, the factors which govern the optimum effective temperature for comfort.

7. Attempt any *one* part of the following: 10 x 1 = 10

(a) Define the function of Cooling Tower, Classify and Explain with neat sketches.

(b) Explain the working of ice manufacturing in ice manufacturing plant.

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